AD7031 Optimization Under Uncertainty

Homework 1 — Solutions

Due Sep 26, start of class

Problem 1. From words to a model **(40 points)**

A transport aircraft is being loaded for a resupply flight. Two cargo types, with value, weight and volume given per pallet:

	ammunition	fuel
value (pts)	4	5
weight (t)	3	2
volume (m^3)	1	3

The aircraft carries at most 60 t and 48 m^3, and orders require that at least 3 pallets of ammunition fly. Cargo is divisible — fractional pallets are allowed. The goal is to maximize total value.

(a) **(8 pts)** Define the decision variables and write the problem as a scalar LP — objective and constraints, one line each, with a few words saying what each constraint enforces.

Solution. Let x_a be the number of ammunition pallets loaded and x_f the number of fuel pallets, both real numbers.

$$\begin{aligned} \max \quad & 4x_a + 5x_f && \text{total value} \\ \text{s.t.} \quad & 3x_a + 2x_f \le 60 && \text{weight limit (t)} \\ & x_a + 3x_f \le 48 && \text{volume limit } (m^3) \\ & x_a \ge 3 && \text{minimum ammunition order} \\ & x_a, x_f \ge 0 && \text{no negative cargo} \end{aligned}$$

(b) **(8 pts)** Now put that LP in the Week 2 canonical vector form $\max \boldsymbol{c}^\top \boldsymbol{x}$ s.t. $A\boldsymbol{x} \le \boldsymbol{b}$: write out $\boldsymbol{c}$, A, $\boldsymbol{b}$ with numbers in them, and report the dimensions of A.

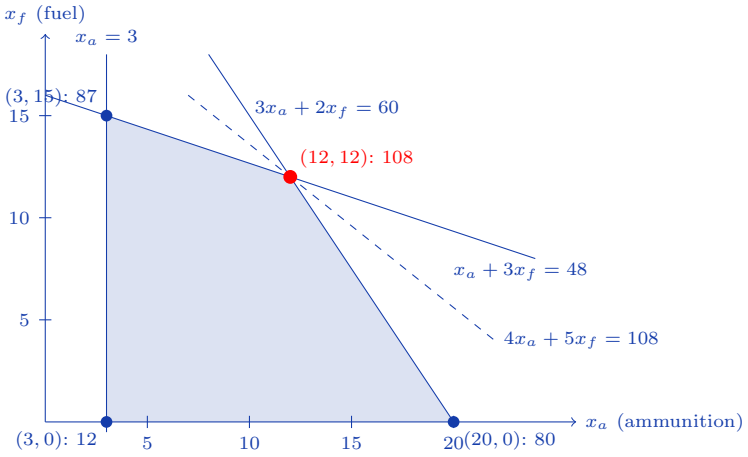
Solution. Take $\boldsymbol{x} = (x_a, x_f)^\top$. The order $x_a \ge 3$ becomes $-x_a \le -3$, and $x_f \ge 0$ becomes $-x_f \le 0$. The row $x_a \ge 0$ is implied by $x_a \ge 3$, so it can be dropped:

$$\boldsymbol{c} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}, \qquad A = \begin{pmatrix} 3 & 2 \\ 1 & 3 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 60 \\ 48 \\ -3 \\ 0 \end{pmatrix} \begin{matrix} \text{weight} \\ \text{volume} \\ x_a \ge 3 \\ x_f \ge 0 \end{matrix}$$

A is 4×2: four constraints, two variables. Keeping the redundant row $-x_a \le 0$ as well gives a 5×2 matrix that describes the same feasible set; either answer is fine, as long as the dimension reported matches the matrix written.

(c) **(14 pts)** Solve it by picture. Draw the feasible region in Desmos and include a screenshot. Then list every vertex with its coordinates and its objective value, and report the optimal solution and its optimal value.

Solution. In Desmos, type the four inequalities as they stand, with $x = x_a$ and $y = x_f$: `3x+2y<=60`, `x+3y<=48`, `x>=3`, `y>=0`. The region where all four shadings overlap is the feasible set — the quadrilateral below.



The four vertices, each with $4x_a + 5x_f$:

vertex	which two constraints meet there	value
$(3, 0)$	$x_a = 3$ and $x_f = 0$	12
$(20, 0)$	weight and $x_f = 0$	80
$(12, 12)$	weight and volume	**108**
$(3, 15)$	volume and $x_a = 3$	87

The optimum is $x_a^\star = 12$, $x_f^\star = 12$ with value 108. The dashed line is the objective level set through the optimum: pushing it any further up and to the right leaves the region, which is why the optimum sits at that corner.

(d) **(10 pts)** Implement the LP with `gurobipy` and solve it. Include a screenshot of your code and its printed output. Check that Gurobi agrees with (c).

Solution.

```
import gurobipy as gp
from gurobipy import GRB

model = gp.Model("cargo")
x_ammo = model.addVar(vtype=GRB.CONTINUOUS, name="x_ammo")   # lower bound 0 by default
x_fuel = model.addVar(vtype=GRB.CONTINUOUS, name="x_fuel")
model.setObjective(4 * x_ammo + 5 * x_fuel, GRB.MAXIMIZE)
model.addConstr(3 * x_ammo + 2 * x_fuel <= 60, name="weight")
model.addConstr(x_ammo + 3 * x_fuel <= 48, name="volume")
model.addConstr(x_ammo >= 3, name="min_ammo")
model.optimize()

print(f"optimal solution ammunition is {x_ammo.X}")
print(f"optimal solution fuel is {x_fuel.X}")
print(f"optimal objective value is {model.ObjVal}")
```

Output:

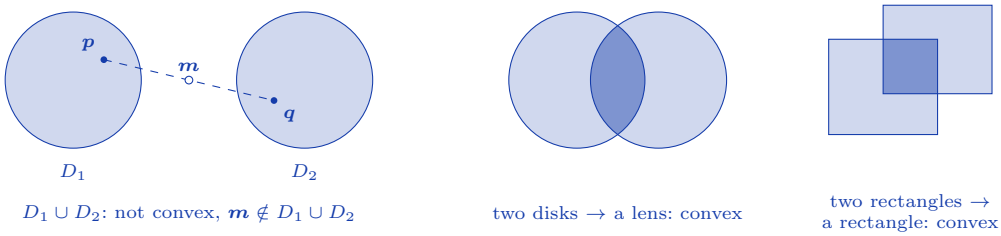
```
optimal solution ammunition is 12.0
optimal solution fuel is 12.0
optimal objective value is 108.0
```

Gurobi returns $(12, 12)$ with value 108, the same corner as the drawing in (c).

Problem 2. Convex sets **(20 points)**

(a) **(10 pts)** Draw two convex sets in $\mathbb{R}^2$ whose *union* is not convex, marking two points and the point on their segment that escapes. Then *try* to draw two convex sets whose *intersection* is not convex.

Solution.



Left: D_1 and D_2 are convex, but $\boldsymbol{p} \in D_1$, $\boldsymbol{q} \in D_2$, and the midpoint $\boldsymbol{m}$ of their segment lies in the gap between them, so the union fails the segment test. Right: every attempt at a non-convex intersection fails — two disks meet in a lens, two rectangles in a rectangle, both convex. This is not bad luck: the intersection of convex sets is always convex (a segment that stays inside each set stays inside their common part).

(b) **(10 pts)** Show that a polyhedron $P = \{\, \boldsymbol{x} \in \mathbb{R}^n : A\boldsymbol{x} \le \boldsymbol{b} \,\}$ is a convex set.

Solution. Take $\boldsymbol{x}, \boldsymbol{y} \in P$, so $A\boldsymbol{x} \le \boldsymbol{b}$ and $A\boldsymbol{y} \le \boldsymbol{b}$, and $\lambda \in [0,1]$. Then

$$A\big(\lambda\boldsymbol{x} + (1-\lambda)\boldsymbol{y}\big) = \lambda\, A\boldsymbol{x} + (1-\lambda)\, A\boldsymbol{y} \;\le\; \lambda\,\boldsymbol{b} + (1-\lambda)\,\boldsymbol{b} = \boldsymbol{b},$$

where the inequality holds row by row because $\lambda \ge 0$ and $1-\lambda \ge 0$: multiplying an inequality by a nonnegative number keeps its direction, and adding two inequalities of the same direction is allowed. So $\lambda\boldsymbol{x} + (1-\lambda)\boldsymbol{y} \in P$, and P passes the segment test.

Problem 3. Convex functions **(20 points)**

(a) **(10 pts)** Suppose $f_k : \mathbb{R}^N \to \mathbb{R}$, $k \in [K]$, are convex. Show that $g(\boldsymbol{x}) = \sum_k f_k(\boldsymbol{x})$ and $g(\boldsymbol{x}) = \max_k f_k(\boldsymbol{x})$ are convex.

Solution. Fix $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^N$ and $\lambda \in [0,1]$, and write $\boldsymbol{z} = \lambda\boldsymbol{x} + (1-\lambda)\boldsymbol{y}$.

Sum. Apply the definition to each f_k and add the K inequalities:

$$g(\boldsymbol{z}) = \sum_k f_k(\boldsymbol{z}) \;\le\; \sum_k \big[\lambda f_k(\boldsymbol{x}) + (1-\lambda) f_k(\boldsymbol{y})\big] = \lambda \sum_k f_k(\boldsymbol{x}) + (1-\lambda)\sum_k f_k(\boldsymbol{y}) = \lambda g(\boldsymbol{x}) + (1-\lambda) g(\boldsymbol{y}).$$

Max. For every single k,

$$f_k(\boldsymbol{z}) \;\le\; \lambda f_k(\boldsymbol{x}) + (1-\lambda) f_k(\boldsymbol{y}) \;\le\; \lambda \max_j f_j(\boldsymbol{x}) + (1-\lambda)\max_j f_j(\boldsymbol{y}) = \lambda g(\boldsymbol{x}) + (1-\lambda) g(\boldsymbol{y}),$$

because replacing $f_k(\boldsymbol{x})$ by the largest of the $f_j(\boldsymbol{x})$ can only increase the right-hand side. The right-hand side does not depend on k, so taking the max over k on the left gives $g(\boldsymbol{z}) \le \lambda g(\boldsymbol{x}) + (1-\lambda) g(\boldsymbol{y})$.

(b) **(5 pts)** Show that $f(x) = x^2$ is convex.

Solution. For $x, y \in \mathbb{R}$ and $\lambda \in [0,1]$, expand the difference between the two sides of the definition:

$$\lambda x^2 + (1-\lambda)y^2 - \big(\lambda x + (1-\lambda)y\big)^2 = \lambda(1-\lambda)\, x^2 - 2\lambda(1-\lambda)\, xy + \lambda(1-\lambda)\, y^2 = \lambda(1-\lambda)\,(x-y)^2 \;\ge\; 0,$$

since $\lambda \ge 0$, $1-\lambda \ge 0$ and a square is nonnegative. Hence $(\lambda x + (1-\lambda)y)^2 \le \lambda x^2 + (1-\lambda)y^2$. (Writing "=" here is the common slip: the two sides are equal only when $x = y$.)

(c) **(5 pts)** Show that $f(\boldsymbol{x}) = \|\boldsymbol{x}\|_2^2$ is convex.

Solution. $\|\boldsymbol{x}\|_2^2 = x_1^2 + x_2^2 + \cdots + x_N^2$ is a sum of the N functions $f_i(\boldsymbol{x}) = x_i^2$. Each f_i is convex by (b) applied to the i-th coordinates: $(\lambda x_i + (1-\lambda)y_i)^2 \le \lambda x_i^2 + (1-\lambda)y_i^2$. A sum of convex functions is convex by (a). So $\|\boldsymbol{x}\|_2^2$ is convex.

Problem 4. Markowitz portfolio **(20 points)**

Three funds — bond, tech, energy — with mean vector $\boldsymbol{\mu}$ and covariance matrix Σ:

$$\boldsymbol{\mu} = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0.000 \\ 0.001 & 0.040 & 0.015 \\ 0.000 & 0.015 & 0.090 \end{pmatrix}, \qquad \begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^3} \quad & \boldsymbol{x}^\top \Sigma \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{\mu}^\top \boldsymbol{x} \ge r, \quad \mathbf{1}^\top \boldsymbol{x} = 1, \quad \boldsymbol{x} \ge \mathbf{0}. \end{aligned}$$

(a) **(15 pts)** Choose your own minimum mean return r, implement the model with `gurobipy`, and print the result. Is your problem feasible? If it is, raise r until it stops being feasible, report that r, and explain why — in plain Korean.

Solution. With $r = 0.10$ (any choice is fine; what matters is the reasoning that follows):

```
import gurobipy as gp
from gurobipy import GRB

model = gp.Model("markowitz")
x_bond   = model.addVar(vtype=GRB.CONTINUOUS, name="x_bond")
x_tech   = model.addVar(vtype=GRB.CONTINUOUS, name="x_tech")
x_energy = model.addVar(vtype=GRB.CONTINUOUS, name="x_energy")

# objective -- portfolio variance x' Sigma x, written out term by term
model.setObjective(
    0.010 * x_bond * x_bond
    + 0.040 * x_tech * x_tech
    + 0.090 * x_energy * x_energy
    + 2 * 0.001 * x_bond * x_tech
    + 2 * 0.015 * x_tech * x_energy,
    GRB.MINIMIZE,
)
r = 0.10
model.addConstr(0.06 * x_bond + 0.10 * x_tech + 0.15 * x_energy >= r, name="target_return")
model.addConstr(x_bond + x_tech + x_energy == 1, name="budget")   # the whole fund is invested
model.optimize()

print(f"optimal solution bond is {x_bond.X}")
print(f"optimal solution tech is {x_tech.X}")
print(f"optimal solution energy is {x_energy.X}")
print(f"optimal objective value is {model.ObjVal}")
```

Output:

```
optimal solution bond is 0.3922872131614318
optimal solution tech is 0.2938830159484385
optimal solution energy is 0.31382977089013236
optimal objective value is 0.016855053202895175
```

So $r = 0.10$ is **feasible**: about 39% bond, 29% tech, 31% energy, with variance 0.0169 (standard deviation ≈ 0.13). Note the budget row is `== 1`, as the model states; the Week 3 notebook's `<= 1` is a different model (the leftover sits in cash) and gives a different portfolio for the same r.

Raising r and re-solving:

r	bond	tech	energy	variance	
0.10	0.392	0.294	0.314	0.0169	feasible
0.12	0.116	0.392	0.493	0.0340	feasible
0.14	0	0.200	0.800	0.0640	feasible
0.15	0	0	1	0.0900	feasible — everything in energy
0.151		—		—	**infeasible** (Gurobi: `Model is infeasible`)

The problem stops being feasible as soon as r exceeds 0.15: $r = 0.15$ is the last feasible value, and $r = 0.151$ (or any $r > 0.15$) has no solution.

Why, in plain Korean. 돈을 세 펀드에 어떻게 나누어 넣더라도, 포트폴리오 전체의 기대수익률은 세 펀드 중 기대수익률이 가장 높은 에너지 펀드의 15%를 넘을 수 없습니다. 가장 좋은 펀드 하나에 전부 넣는 것이 기대수익률을 가장 높이는 방법이고, 그때가 딱 15%이기 때문입니다. 그래서 요구 수익률 r이 15%를 넘는 순간, 어떤 배분도 조건을 만족하지 못해 문제가 풀리지 않습니다 (infeasible). 반대로 r이 작을수록 (예: 0.01) 조건은 느슨해지고, 답은 변동성이 가장 작은 배분으로 갑니다.

(b) **(5 pts)** Is the Markowitz problem a convex problem? If it is, explain why.

Solution. Yes. *Objective.* $f(\boldsymbol{x}) = \boldsymbol{x}^\top \Sigma \boldsymbol{x}$ is convex because Σ is a covariance matrix, hence positive semidefinite ($\boldsymbol{v}^\top \Sigma \boldsymbol{v} \geq 0$ for every $\boldsymbol{v}$ — it is the variance of the portfolio $\boldsymbol{v}$). Repeating the computation

of Problem 3(b) with vectors,

$$\lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}) - f\big(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}\big) = \lambda(1-\lambda)\,(\boldsymbol{x}-\boldsymbol{y})^{\top} \Sigma\,(\boldsymbol{x}-\boldsymbol{y}) \;\geq\; 0.$$

Feasible set. $\{\, \boldsymbol{x} : \boldsymbol{\mu}^{\top}\boldsymbol{x} \geq r,\ \mathbf{1}^{\top}\boldsymbol{x} = 1,\ \boldsymbol{x} \geq \mathbf{0} \,\}$ is described by linear inequalities and one linear equality ($= 1$ is ≤ 1 and ≥ 1 together), so it is a polyhedron, which is convex by Problem 2(b). A convex objective over a convex feasible set is a convex problem — which is why Gurobi finds the global optimum without any search.